

Homework 1

2026-05-04

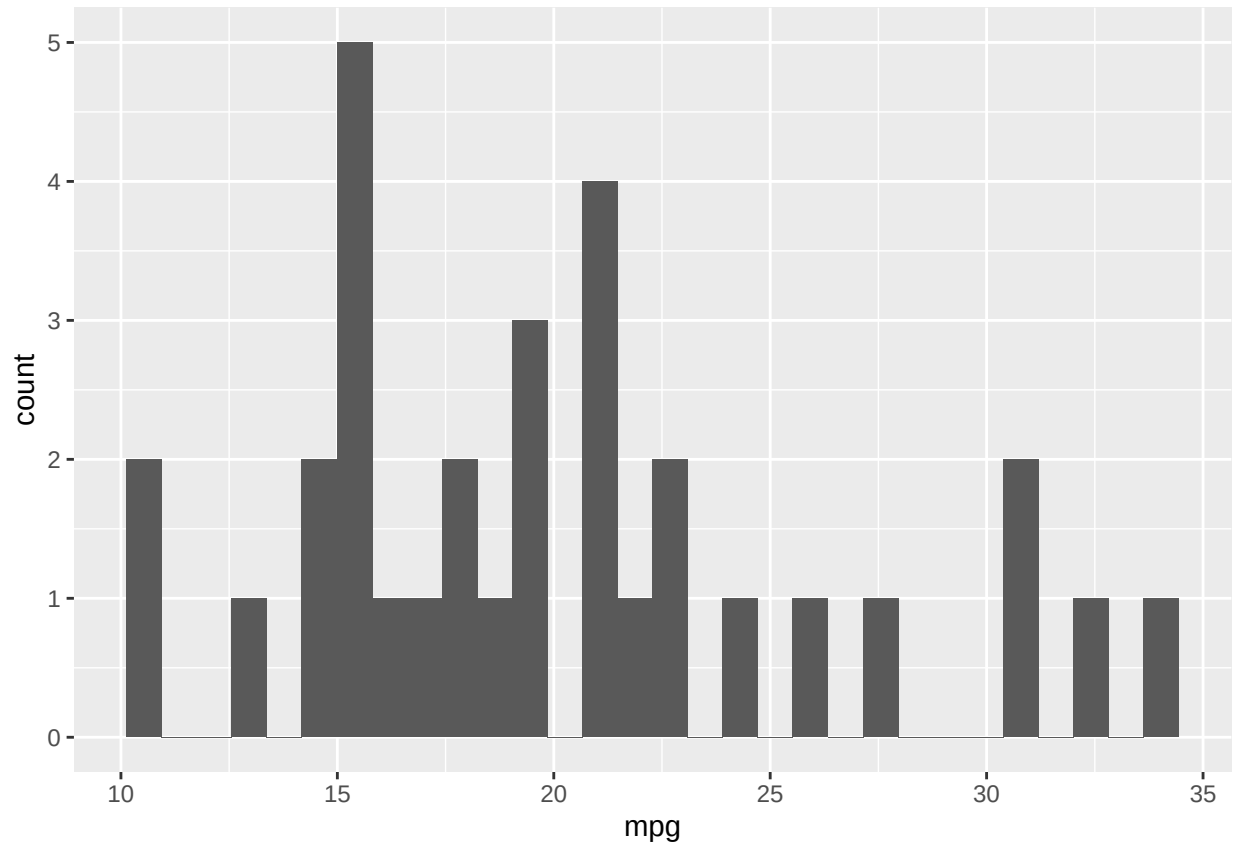
Replicate the following plots using the `ggplot2` package. Submit your code and the plots as a single PDF file (like this one, but without hiding the R code) on Moodle!

1. Check the structure of the `mtcars` dataset!

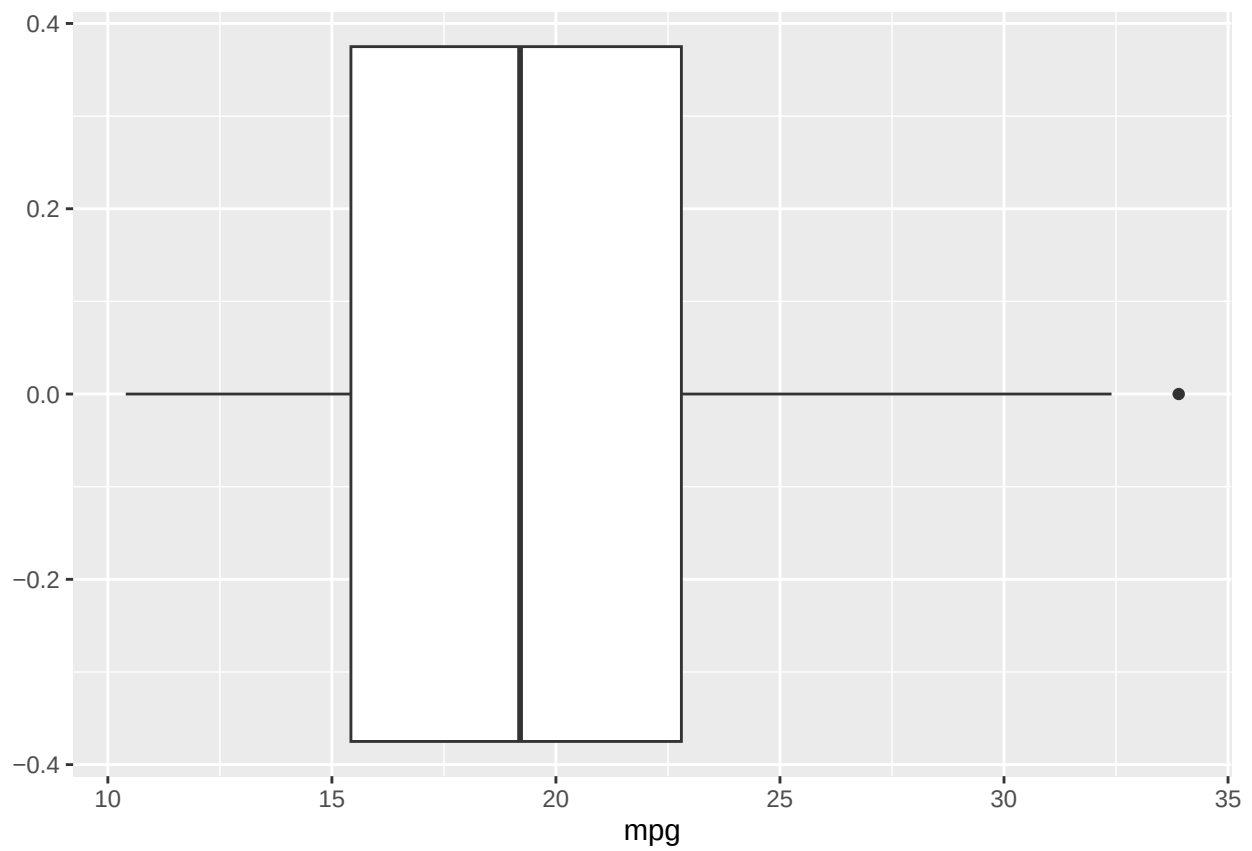
```
## 'data.frame':   32 obs. of  11 variables:
## $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
## $ cyl : num   6  6  4  6  8  6  8  4  4  6 ...
## $ disp: num  160 160 108 258 360 ...
## $ hp  : num  110 110  93 110 175 105 245  62  95 123 ...
## $ drat: num   3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
## $ wt  : num   2.62 2.88 2.32 3.21 3.44 ...
## $ qsec: num  16.5 17 18.6 19.4 17 ...
## $ vs  : num   0  0  1  1  0  1  0  1  1  1 ...
## $ am  : num   1  1  1  0  0  0  0  0  0  0 ...
## $ gear: num   4  4  4  3  3  3  3  4  4  4 ...
## $ carb: num   4  4  1  1  2  1  4  2  2  4 ...
```

2. Create a histogram of the mpg column!

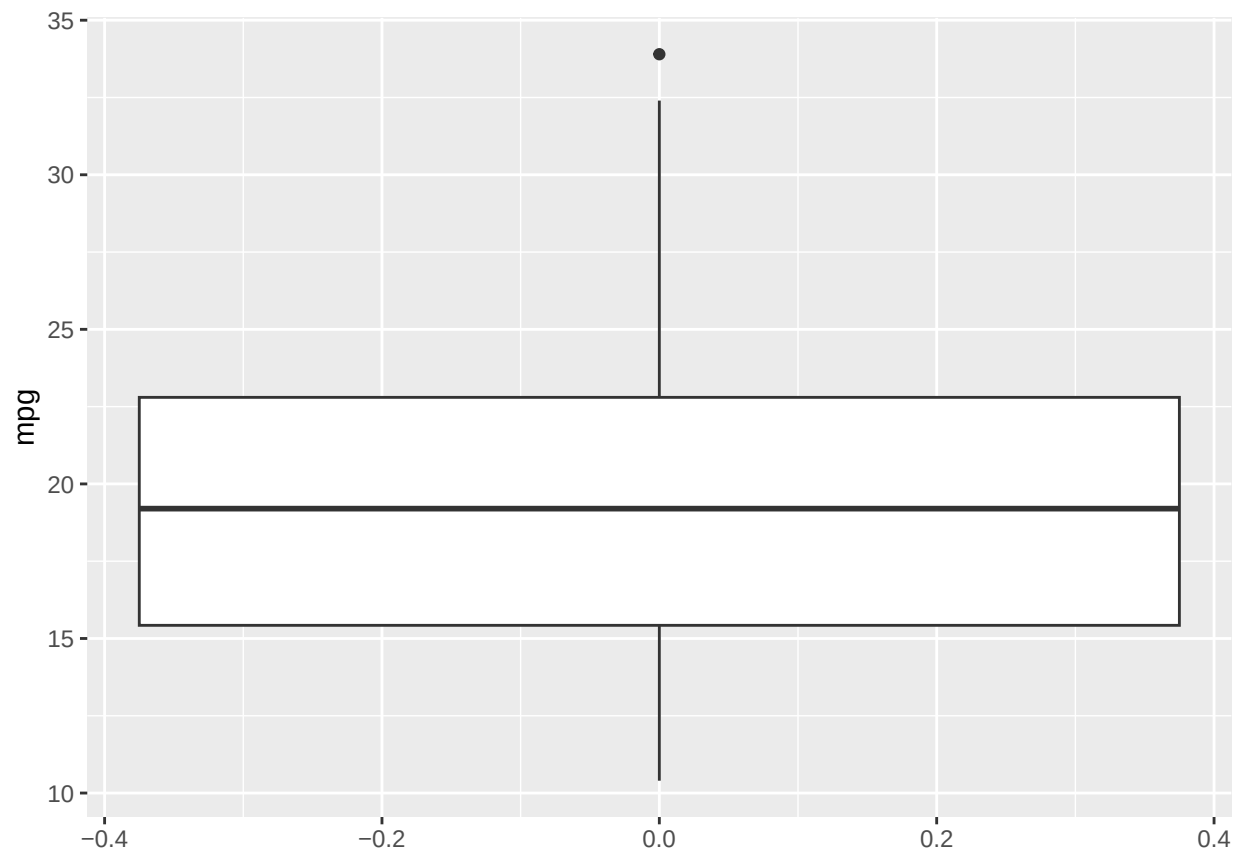
```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



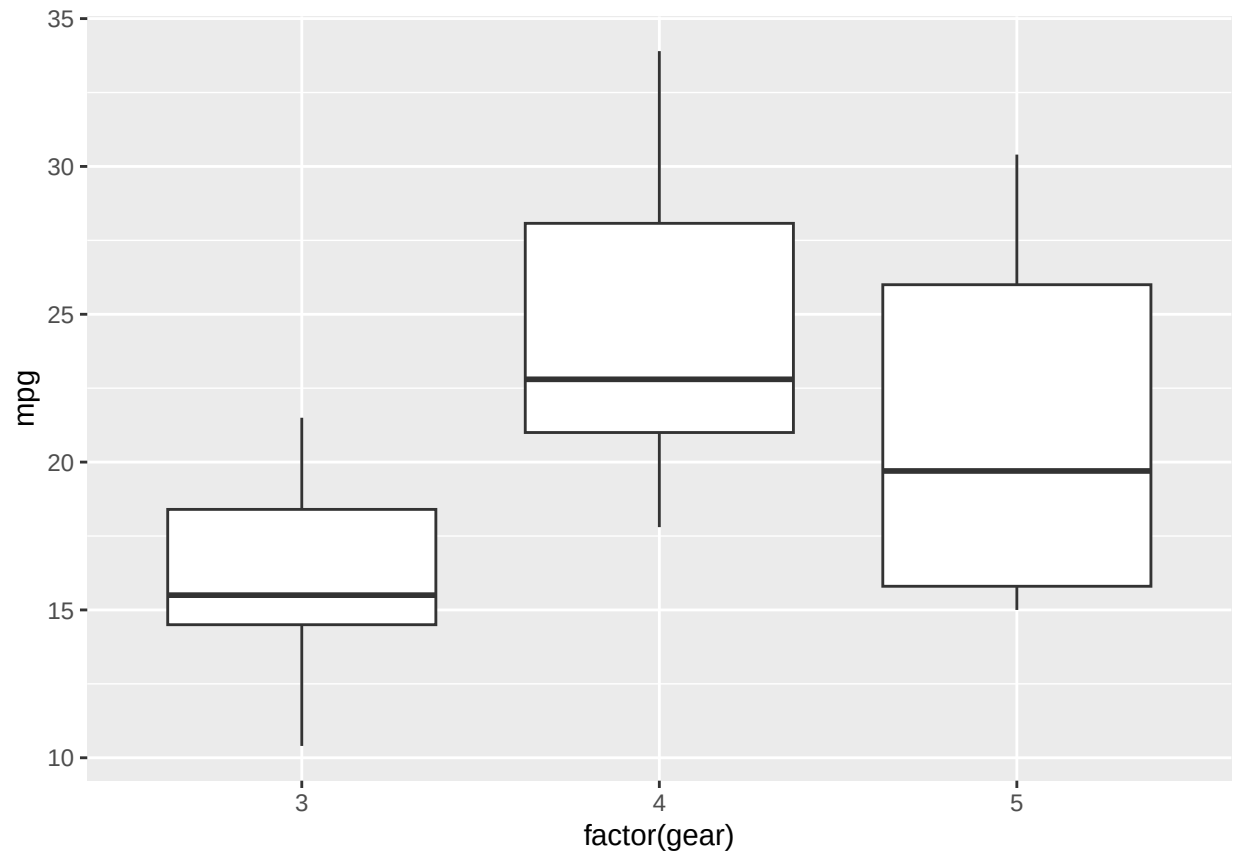
3. Create a horizontal boxplot of the mpg column!



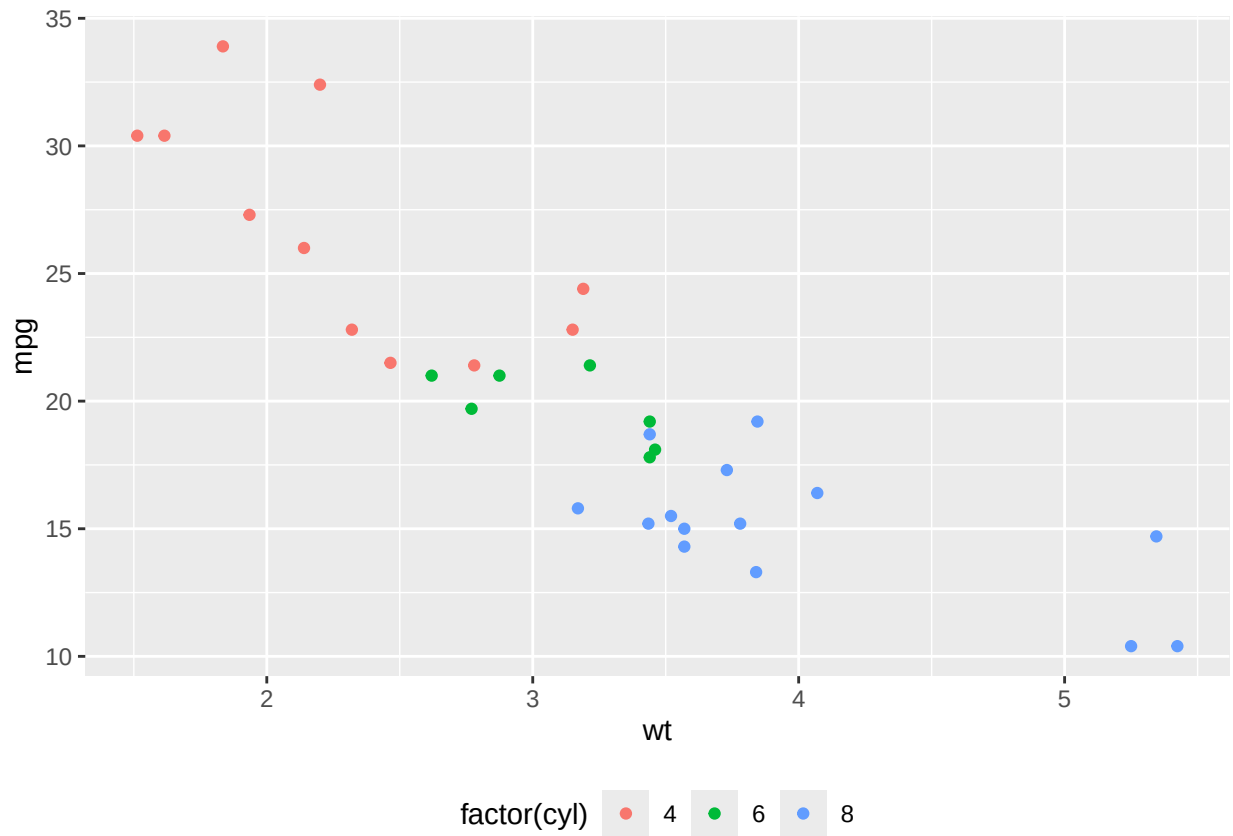
4. Create a vertical boxplot of the mpg column:



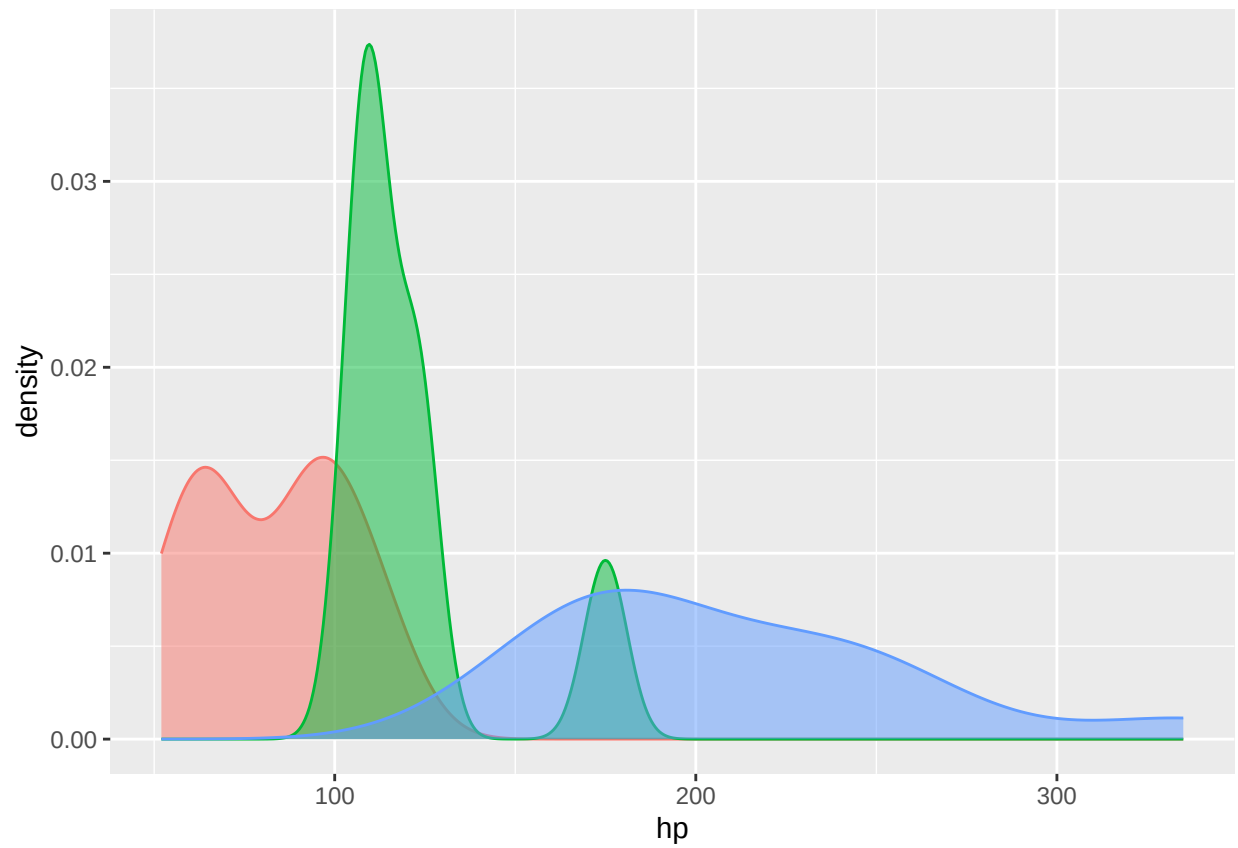
5. Create a boxplot on the `mpg` column split by the number of gears! Note the x axis label as a hint for converting the continuous variable to discrete.



6. Create a scatter plot of the `mpg` column against the `wt` column, color the points by the number of cylinders! Move the legend to the bottom of the plot using the `theme` function.

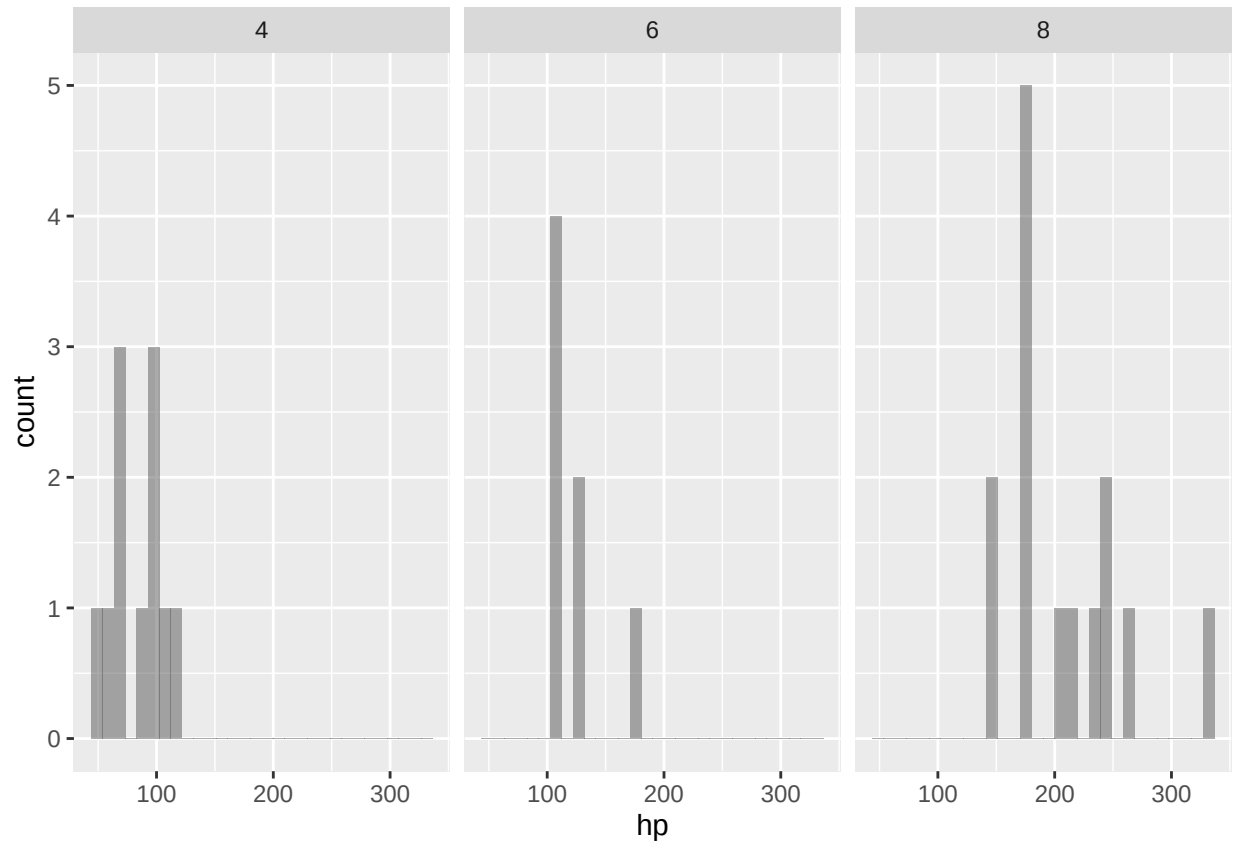


7. Create a semi-transparent density plot on the `hp` column split (colored) by the number of cylinders! Note that there's no legend, and the outline of the density plot is the same color as the fill.



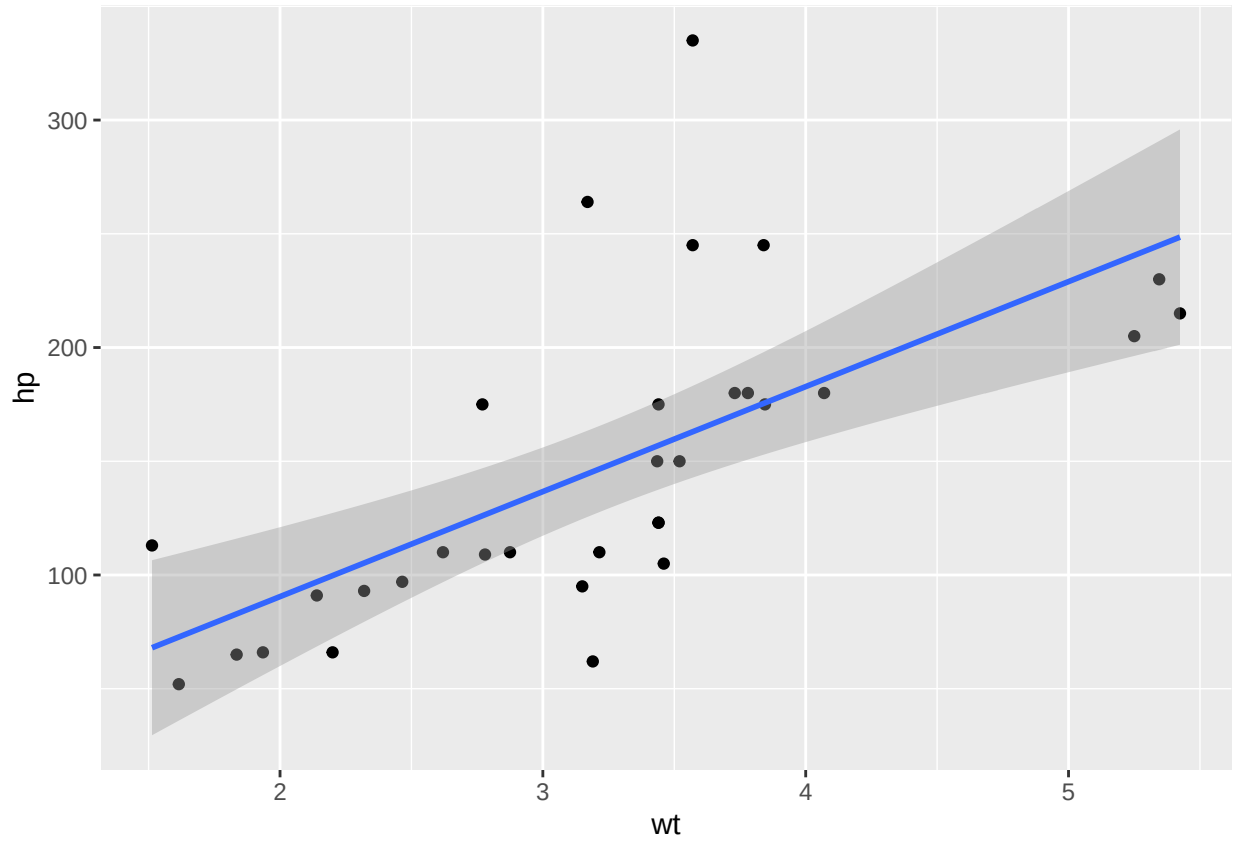
8. Create histograms on `hp` split by the number of cylinders using the `facet_wrap` function!

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



9. Create a scatter plot on hp against wt with a linear regression line!

```
## `geom_smooth()` using formula = 'y ~ x'
```



10. Replicate the previous plot, but color the points by the number of cylinders! Keep the regression line the same color as the points.

```
## `geom_smooth()` using formula = 'y ~ x'
```

